Syllabus
Software Engineering Department

Course Name		
Course No.	3503836– Computer Networks	
Lecturers' Name	Itzhak Nudler	nudler@shenkar.ac.il
Teaching Assistant		
Year	2026	תשפ"ו
Weekly Hours	4 Lecture Hours	
Credits	4	
Prerequisite	Computer Architecture Object Oriented Programming Data Structures	

Course Summary
The course introduces the principles and practice of computer networking, with emphasis on the Internet. The course focus on the TCP/IP protocol suite. The structure and components of computer networks, packet switching, layered architectures, TCP/IP, communication devices such as organization and home routers, physical communication channels, error control, window flow control, local area networks (Ethernet, WiFi),

LEARNING METHODS	- Frontal lectures - Self readings - Qa and As sessions at class - Home exercises

Learning Outcomes
1. Become a more skilled programmer in regard with the use of computer networks, including the development of network applications (e.g. web appliactions). By understanding the sockets interface, and in particular the important details 'behind' this interface. 2. Know and understand the main components of networks applications and of computer networks and the considerations in their design. 3. Understand how ro desifn good network applications: coorect, high-performance and scalable. 4. Understand how do design correct and efficient network protocols.

Course Requirements and Grading			
Course Requirements	At least 60 in the final exam		
Grading	Assignments	5%	
	Test	95%	

Class Attendance	Mandatory. According to Shenkar's regulations

Course Plan- List of Topics		
	Topic	Reading/Assignment
1	Extended Introduction: - communication networks in general and computer networks in particular - The Internet - Communication protocols	"Computer Networking: A Top-Down Approach Featuring the Internet - 8th Edition", Kurose and Ross, Chapter 1
2	- layering - network architecture - The TCP/IP network modle - the 7 layers model (OSI) vs. the TCP/IP model	"Computer Networking: A Top-Down Approach Featuring the Internet - 8th Edition", Kurose and Ross, Chapter 1
3	The application layer	"Computer Networking: A Top-Down Approach Featuring the Internet - 8th Edition", Kurose and Ross, Chapter 2
4	The sockets interface API	"Beej's Guide to Network Programming - Using Internet Sockets"
5	The sockets interface: 'behind' the API	
6	The transport layer: TCP, UDP, TCP vs/ UDP	"Computer Networking: A Top-Down Approach Featuring the Internet - 8th Edition", Kurose and Ross, Chapter 3
7	- Reliable Data Transfer (RDT) mechanisms – a general discussion - TCP's RDT mechanisms.	"Computer Networking: A Top-Down Approach Featuring the Internet - 8th Edition", Kurose and Ross, Chapter 3
8	- Congestion control – a general discussion. - TCP's congestion control mechanisms.	"Computer Networking: A Top-Down Approach Featuring the Internet - 8th Edition", Kurose and Ross, Chapter 3
9	The network layer: - internetworking. - switching and routing - the IP datagram format	"Computer Networking: A Top-Down Approach Featuring the Internet - 8th Edition", Kurose and Ross, Chapter 4
10	IP: - IP addresses - global vs. local addresses - global addresses assignment	"Computer Networking: A Top-Down Approach Featuring the Internet - 8th Edition", Kurose and Ross, Chapter 4

	schemes - IPv6	
11	Routing: - static and dynamic routing - Routing protocols IP and Routing	"Computer Networking: A Top-Down Approach Featuring the Internet - 8th Edition", Kurose and Ross, Chapter 5

Bibliography
"Computer Networking: A Top-Down Approach Featuring the Internet - 8th Edition", James F. Kurose and Keith W. Ross, Addison-Wesley. 2021
"Computer Network – 5th Edition", A. S. Tanenbaum. Prentice-Hall. 2010
"Computer Networks: A Systems Approach - 5th Edition", Larry L. Peterson , Bruce S. Davie, The Morgan Kaufmann Series in Networking, 2011.
"Beej's Guide to Network Programming - Using Internet Sockets". 2022 Available online at: https://beej.us/guide/bgnet/html/split/